



CPSC 100

Computational Thinking

Computational Thinking and Markdown

Instructor: Dr. Firas Moosvi
Department of Computer Science
University of British Columbia

Agenda

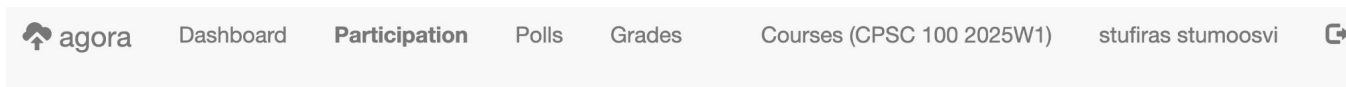
- **Recap:** Computational Thinking
- Class discussions using Agora
- **Debrief:** Class Activity on Card Sorting
- Introduction to Markdown

Class Discussion Using Agora



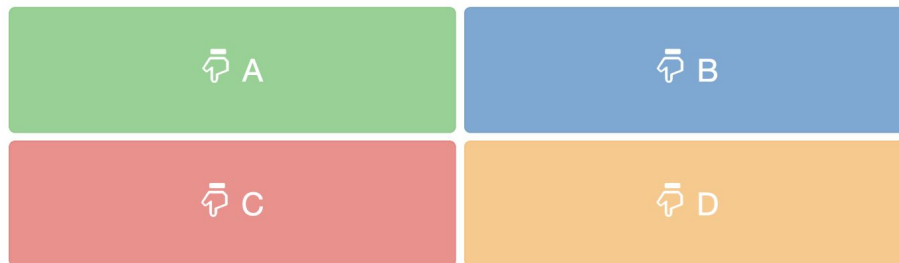
Agora: Discussions and Polling

This is what we will use during class to answer poll questions and have discussions!



CPSC 100 2025W1

Participation Points: + (Phantom)



Message Board

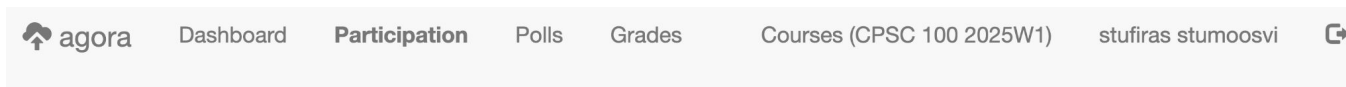
Lecture not started.



Agora: Discussions and Polling

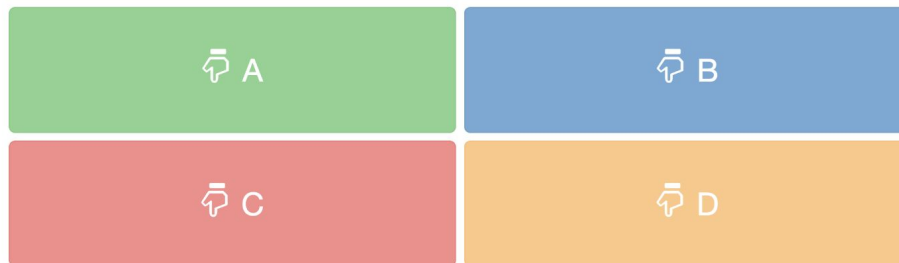
Page: agora.students.cs.ubc.ca (login with your CWL)

Enroll code: **conformitygate**



CPSC 100

Participation Points: + (Phantom)



Message Board

Lecture not started.



Clicker Question

What year of University are you in?

- a) 1st year
- b) 2nd year
- c) 3rd year
- d) 4th year
- e) 5th year or higher



Participation Question

Why do people today live longer today than in pre-historic times?

Fewer Wars

Science

Higher
Intelligence

Better Mental
Health



Computational Thinking

What is Computational Thinking?

“Computational thinking is the **thought processes** involved in **formulating problems and their solutions** so that the solutions are in a form that can be **effectively** carried out by an **information-processing agent**”

[Cuny, Snyder, Wing]

What is Computational Thinking?

“Computational thinking is the **thought processes** involved in **formulating problems and their solutions** so that the solutions are in a form that can be **effectively** carried out by an **information-processing agent**”

[Cuny, Snyder, Wing]

What is Computational Thinking?

“Computational thinking is the **thought processes** involved in **formulating problems and their solutions** so that the solutions are in a form that can be **effectively** carried out by an **information-processing agent**”

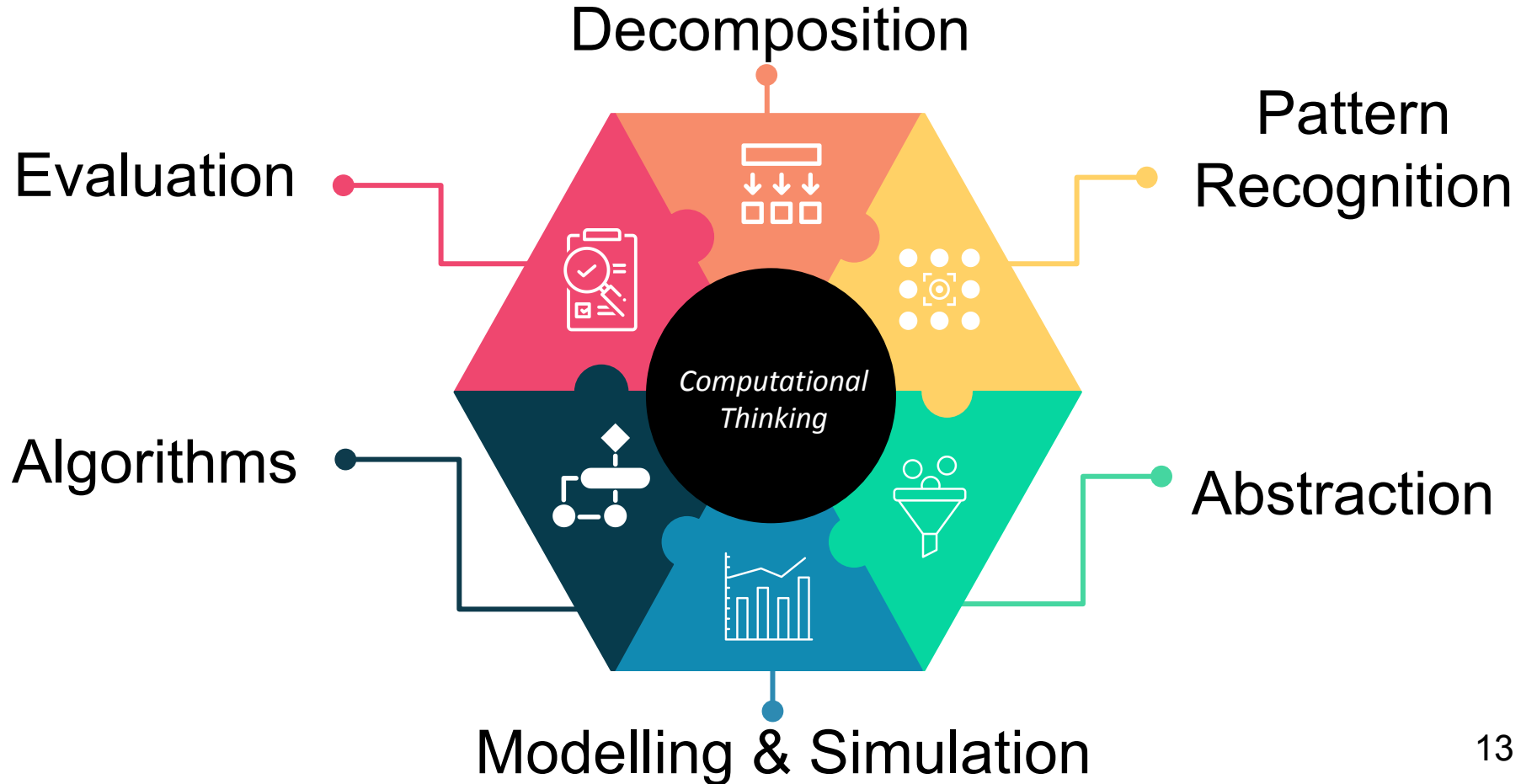
[Cuny, Snyder, Wing]

What is Computational Thinking?

“Computational thinking is the **thought processes** involved in **formulating problems and their solutions** so that the solutions are in a form that can be **effectively** carried out by an **information-processing agent**”

[Cuny, Snyder, Wing]

Computational Thinking Skills





DECOMPOSITION

Breaking down problems into smaller, easier parts.



PATTERN RECOGNITION

Using patterns in information to solve problems.



ABSTRACTION

Finding information that is useful and taking away any information that is unhelpful.



MODELLING AND SIMULATION

Trying out different solutions or tracing the path of information to solve problems.



ALGORITHMS

Creating a set of instructions for solving a problem or completing a task.



EVALUATION

Assessing a solution to a problem and using that information again on new problems.

Debrief: Sorting Class Activity



Class Activity: Sort the Cards

Imagine a robot must arrange a set of cards in ascending order (Ace to King, Same suit).

The robot can only follow your instructions.

Task [Groups of 3-4]

Create a clear set of steps/instructions to sort the card



Participation Question

Did you come up with a list of steps to sort the cards?

Successful: What worked?

Unsuccessful: Where did it fail ?

Logistics

Counter Example



Introduction to Markdown



What is a Markup Language? #10



[Redacted]

22 hours ago in **Content Support**



PIN



STAR



WATCH

118

VIEWS



Question Number

What is the lecture number/section, question number, label, or identifier ?

For example: "Lecture 2A section on Absolute/Relative Paths"

Lab 0



Question Text

Please include the full question text, as a copy/paste or as a screenshot.

What is the definition of a markup language? I am new to coding and do not really understand the principle of a markup language. Why do they exist? What is the difference between a "markup" vs other languages? Why are we learning this first, is this significant somehow?



What problem are you having and what have you tried so far?

Describe the problem in detail and include any relevant screenshots, error messages and small snippets of code as needed.

I have tried searching this up on the internet, but it has not answered my questions.

[Comment](#) [Edit](#) [Delete](#) [Endorse](#) ...



Firas Moosvi **INSTRUCTOR**

16 hours ago



Great questions! Let me try answering them:



What is the definition of a markup language?

A markup language is a way to add formatting and structure to plain text. For example, "marking up" text can include anything from making text **bold**, *italics*, appear larger in font size, change the font colour, etc...

Think about an essay that you submit for a class - your grader will then "mark up" the text by adding circles, underlines, highlights, and other annotations. This is just another example of "markup".

The most famous example of a Markup language is "HTML" - that's what makes up all the websites on the internet. HTML stands for hypertext markup language.

Why do [markup languages] exist?

They exist because people always want to add formatting and markup to text to spice it up! It's also a way of representing textual data that is rich and dynamic, and allows for people to read content that's formatted instead of a wall of text.

What is the difference between a "markup" vs other languages?

In the computer science world, there are other types of languages as well. For example, "programming" language, "scripting language", "query language", "assembly language" etc...

You don't really need to know what the other languages are or do, but briefly, you can think of programming (or scripting) languages as ones that provide instructions to a computer on what it needs to do whereas a markup language just dictates how text is displayed or presented.

Why are we learning this first, is this significant somehow?

Very insightful question! We are learning how to use a markup language (markdown) so that we can use it in this course on the class activities, learning logs, and eventually, on tests.

Reference: Geeks for Geeks

Difference Between Programming, Scripting, and Markup Languages

Last Updated : 11 Jul, 2025



When it comes to making a website or app, coding involves basically **three types of languages**, i.e the **programming language**, **Scripting Language** and **Markup Language**. There are a lot of people who consider coding as just developing or making a website, but they need to understand that every single language fits into a particular category, and we need to know which category that language fits into.



Markdown Syntax - Headings

Headings

To create a heading, add number signs (#) in front of a word or phrase. The number of number signs you use should correspond to the heading level. For example, to create a heading level three (<h3>), use three number signs (e.g., ### My Header).

Markdown	HTML	Rendered Output
# Heading level 1	Heading level 1	Heading level 1
## Heading level 2	Heading level 2	Heading level 2
### Heading level 3	Heading level 3	Heading level 3
#### Heading level 4	Heading level 4	Heading level 4
##### Heading level 5	Heading level 5	Heading level 5
##### Heading level 6	Heading level 6	Heading level 6



Resources for Learning Markdown

- [Markdown Cheat Sheet](#)
- [CommonMark Tutorial](#)
 - You will be doing this for Lab 0 this week (on your own)



Markdown Syntax - Bold Text

Bold

To bold text, add two asterisks or underscores before and after a word or phrase. To bold the middle of a word for emphasis, add two asterisks without spaces around the letters.

Markdown	HTML	Rendered Output
I just love <code>**bold text**</code> .	You can learn about HTML on your own (if you want)!	I just love bold text .
I just love <code>__bold text__</code> .		I just love bold text .
Love <code>**is**</code> bold		Love is bold



Markdown Syntax - Italics Text

Italic

To italicize text, add one asterisk or underscore before and after a word or phrase. To italicize the middle of a word for emphasis, add one asterisk without spaces around the letters.

Markdown	HTML	Rendered Output
Italicized text is the <code>*cat's meow*</code> .	<code><i>You can learn about HTML on your own (if you want)!</i></code>	Italicized text is the <i>cat's meow</i> .
Italicized text is the <code>_cat's meow_</code> .		Italicized text is the <i>cat's meow</i> .
<code>A*cat*meow</code>		<i>Acatmeow</i>



Markdown Syntax - Blockquotes

Blockquotes

To create a blockquote, add a > in front of a paragraph.

```
> Dorothy followed her through many of the beautiful rooms in her castle.
```

The rendered output looks like this:

```
Dorothy followed her through many of the beautiful rooms in her castle.
```



Markdown Syntax - Blockquotes

Blockquotes with Multiple Paragraphs

Blockquotes can contain multiple paragraphs. Add a > on the blank lines between the paragraphs.

```
> Dorothy followed her through many of the beautiful rooms in her castle.  
>  
> The Witch bade her clean the pots and kettles and sweep the floor and keep the fire fed with
```

The rendered output looks like this:

Dorothy followed her through many of the beautiful rooms in her castle.

The Witch bade her clean the pots and kettles and sweep the floor and keep the fire fed with wood.



Markdown Syntax - Blockquotes

Nested Blockquotes

Blockquotes can be nested. Add a >> in front of the paragraph you want to nest.

```
> Dorothy followed her through many of the beautiful rooms in her castle.  
>  
>> The Witch bade her clean the pots and kettles and sweep the floor and keep the fire fed with
```

The rendered output looks like this:

Dorothy followed her through many of the beautiful rooms in her castle.

The Witch bade her clean the pots and kettles and sweep the floor and keep the fire fed with wood.



Markdown Syntax - Ordered Lists

Ordered Lists

To create an ordered list, add line items with numbers followed by periods. The numbers don't have to be in numerical order, but the list should start with the number one.

Markdown	HTML	Rendered Output
<pre>1. First item 2. Second item 3. Third item 4. Fourth item</pre>	<pre><ol style="list-style-type: none;"> 1. First item 2. Second item 3. Third item 4. Fourth item </pre>	<pre>1. First item 2. Second item 3. Third item 4. Fourth item</pre>
<pre>1. First item 1. Second item 1. Third item 1. Fourth item</pre>		<pre>1. First item 2. Second item 3. Third item 4. Fourth item</pre>

You can learn
about HTML on
your own (if you
want)!



Markdown Syntax - Ordered Lists

Ordered Lists

To create an ordered list, add line items with numbers followed by periods. The numbers don't have to be in numerical order, but the list should start with the number one.

Markdown	HTML	Rendered Output
<pre>1. First item 2. Second item 3. Third item 1. Indented item 2. Indented item 4. Fourth item</pre>	<pre>You can learn about HTML on your own (if you want)!</pre>	<pre>1. First item 2. Second item 3. Third item 1. Indented item 2. Indented item 4. Fourth item</pre>



Markdown Syntax - Ordered Lists

Unordered Lists

To create an unordered list, add dashes (-), asterisks (*), or plus signs (+) in front of line items. Indent one or more items to create a nested list.

Markdown	HTML	Rendered Output
- First item - Second item - Third item - Fourth item	You can learn about HTML on your own!	• First item • Second item • Third item • Fourth item
* First item * Second item * Third item * Fourth item		• First item • Second item • Third item • Fourth item



Markdown Syntax - Ordered Lists

Unordered Lists

To create an unordered list, add dashes (-), asterisks (*), or plus signs (+) in front of line items. Indent one or more items to create a nested list.

Markdown	HTML	Rendered Output
<pre>- First item - Second item - Third item - Indented item - Indented item - Fourth item</pre>	You can learn about HTML on your own!	• First item • Second item • Third item ◦ Indented item ◦ Indented item • Fourth item



Markdown Syntax - Horizontal Lines

Horizontal Rules

To create a horizontal rule, use three or more asterisks (***), dashes (---), or underscores (___) on a line by themselves.

```
***
```

```
---
```

```
_____
```

The rendered output of all three looks identical:



Markdown Syntax - Links

Links

To create a link, enclose the link text in brackets (e.g., [Duck Duck Go]) and then follow it immediately with the URL in parentheses (e.g., (https://duckduckgo.com)).

```
My favorite search engine is [Duck Duck Go](https://duckduckgo.com).
```

The rendered output looks like this:

My favorite search engine is [Duck Duck Go](https://duckduckgo.com).



Markdown Syntax - Tables

Tables

To add a table, use three or more hyphens (---) to create each column's header, and use pipes (|) to separate each column. For compatibility, you should also add a pipe on either end of the row.

```
| Syntax      | Description |
| -----    | -----    |
| Header      | Title       |
| Paragraph   | Text        |
```

The rendered output looks like this:

Syntax	Description
Header	Title
Paragraph	Text

Markdown Syntax - Images

Images

To add an image, add an exclamation mark (!), followed by alt text in brackets, and the path or URL to the image asset in parentheses. You can optionally add a title in quotation marks after the path or URL.

```
![The San Juan Mountains are beautiful!](/assets/images/san-juan-mountains.jpg "San Juan Mounti
```

The rendered output looks like this:



Class Activity



Class Activity: Sort the Cards

Imagine a robot must arrange a set of cards in ascending order (Ace to King, Same suit).

The robot can only follow your instructions.

Task [Groups of 3-4]

Create a clear set of steps/instructions to sort the card



Participation Question

Did you come up with a list of steps to sort cards?

Successful: What worked?

Unsuccessful: Where did it fail ?

Logistics

Counter Example





Wrap up



Wrap Up

- Markdown Syntax (no need to memorize!)
- Reminder: Do Lab 0 on your own!
 - Labs start next week!
- Remember: Learning Log and Test 0!
- Reminder: Join Ed Discussion board (link on canvas)